DTU

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Why AI Matters Now?

- Increasing pressure on publication and funding
  - “Competitors” use AI big-time
  - Will lead to better written, more compliant competitive applications
- Rapid growth in data and complexity
- Administrative workload and requirements are increasing
- AI offers productivity, insight, and scalability

👉 Key question: How can researchers effectively use AI without compromising quality?

Where AI may help Most?

- Writing & publishing
- Mathematical modelling & data analysis
- Administrative and project tasks

AI for writing

- Language control and proof reading
- Editing documents, e.g. propose how to reduce length of documents, avoiding repetitions
  - Improving clarity and conciseness
  - Explaining complex theories in simpler terms
- Writing abstracts, summaries, review sections
- Checking and suggesting references

AI for literature review

- Literature mapping: Summarizing large volumes of papers and identifies gaps
- Connecting ideas across fields (e.g., transport + behavioral science)
- Cluster literature into themes or categories and structuring review sections
- Identifying contradictions or research gaps

- BUT be aware of possible
  - Hallucinated or incorrect references, wrong interpretation of reviews
  - Lack of scientific rigor

AI in the publication process

- Identifying relevant journals based on abstract or paper draft, and possible recommending rescoping paper
  - Note best to used downloaded trained LLM not to reveal the content of the paper
- Creating submission packages
  - Letter to the editor
  - Formatting to specific journal format and upload requirements,…
  - Help suggesting reviewers
- Responses to reviews
  - Draft response letters to review
  - Strengthening arguments after feedback, and drafting responses to reviewers
  - Rewriting paper sections based on comments

AI in peer review processes

- Has changed review processes for good or bad
  - Journals autogenerate proposals of reviewers who are not always relevant
  - Reviewers use AI to generate reviews (some universities have #reviews as PKIs, publishers has began counting this),…
  - Automation in low-quality journals

AI for mathematical research

- Symbolic Mathematics & Derivations
  - Solving equations (algebraic, differential, integral)
  - Performing symbolic derivations step-by-step
  - Simplifying complex expressions
  - Helping to proof hypothesis or verifying mathematical proofs or steps
- Numerical computations and simulations
- Refine complex model systems
- Pattern discovery and data-driven insights
- Code generation

- BUT be aware of possible
  - Errors in reasoning and mathematics
  - Lack of scientific rigor

AI for data analysis

- Data Interpretation & Explanation
- Detecting and/or cleaning outliers
- Translating results into clear narratives
- Suggesting interpretations of statistical outputs
- Creating and explaining figures, tables, and patterns
- Linking findings to existing literature

Use of AI as a research method

- AI assistance when formulating models, testing and refining model specifications
- Enhancing models predictive power (e.g. model-based ML)
- ML as a learning method within optimization models, when solving bi-level problems, wider range of fixed-point problems (equilibrium in model systems)
- LLMs for processing qualitative data like survey answers, POIs evaluations etc.
- Possible to quantify new type of data sources
- Image recognition and classification
- Virtual reality and simulators

- BUT when using ML and “brute force calculations” one may forget laws from physics, statistics, economics,…

AI in Grant Writing (Pre-Award)

- Help structuring proposals
- Draft administrative sections in proposals (impact, dissemination, project management)
  - Sometime AI knows specific call-requirements (like for DUT, Horizon). Seems better trained for large repetitive calls, than niche calls (like DFF green call)
- Summarizing call requirements
- Ensure compliance with guidelines (both the scientific call text, and other requirements)
  - Our applications often fail on specific ignored requirements, e.g. Marie Curie applications ignoring training and/or secondment aspects
- Funding support; learn patterns from winning and rejected grants, and rejection letters
- Help making presentations based on proposals, e.g. in powerpoint

AI in Project Management (Post-Award)

- Assist with reporting and progress updates
- Classifies and tags research documents and data
- Taking notes and summaries from meetings (e.g. MeetGeek type of tools, etc.)
  - Be aware of GDPR and ethics of this
- Generate data management plan from the application (should be possible?)
- Help identifying issues in grant agreement, consortia agreements
- Help with arguments related to GDPR, legal, etc.
- Reporting & Compliance
  - Keep tracks deadlines and submission requirements
  - Periodic project reports
  - Summarize milestones and deliverables
  - Helps ensuring compliance with funding rules

Administrative Automation

- Helps in recruitment
  - Systematic call distribution
  - Sorting and ranking applications
- Summarize long email threads
- Workflow automation

⚖️ Limitations of AI (AI generated)

- Hallucinated or incorrect references, wrong interpretation of reviews
- Errors in reasoning and mathematics
- Lack of scientific rigor
- Bias in training data
- Ethical concerns (authorship, transparency)

👉 AI requires critical human oversight

🎯 Key Takeaways (AI generated)

- AI may significantly increases research productivity
- Strongest in writing, data analysis, and admin tasks
- Useful for augmentation, not replacement
- Critical thinking remains essential

👉 AI is a co-pilot — not the researcher